

$$\text{increase } \hat{\Omega}_{s2} = \frac{\hat{\Omega}_{p1} \hat{\Omega}_{p2}}{\hat{\Omega}_{s1}}$$

Ex: elliptic bandpass filter

$$\hat{\Omega}_{p1} = 20 \text{ k} \quad \hat{\Omega}_{p2} = 45 \text{ k} \quad \alpha = 0.25 \text{ dB}$$

$$\hat{\Omega}_{s1} = 15 \text{ k} \quad \hat{\Omega}_{s2} = 50 \text{ k} \quad A = 50 \text{ dB}$$

① stopband edge frequency of analog L.P. filter?

$$\hat{\Omega}_{p1} \hat{\Omega}_{p2} = 900 > \quad \hat{\Omega}_{p1} = \frac{\hat{\Omega}_{s1} \hat{\Omega}_{s2}}{\hat{\Omega}_{p2}} = \frac{750}{45} = 16.67 \text{ k}$$

$$\hat{\Omega}_{s1} \hat{\Omega}_{s2} = 750$$

$$\hat{\Omega}_{p1} \hat{\Omega}_{p2} = \hat{\Omega}_{s1} \hat{\Omega}_{s2} \quad \checkmark$$

$$\Omega_p = 1$$

$$\Omega_s = (1) \frac{750 - 225}{15 \cdot 25}$$

$$\Omega_o^2 = \hat{\Omega}_{p1} \hat{\Omega}_{p2} = 750$$

$$\hat{\Omega}_{s1}^2 = 225$$

$$\Omega_s = 1.4 \text{ kHz}$$

$$B_w = \hat{\Omega}_{p2} - \hat{\Omega}_{p1} = 25$$

② Filter order, elliptic?

$$N = \frac{2 \log_{10}(4/k_1)}{\log_{10}(1/\rho)}$$

$$k' = \sqrt{1 - k^2}$$

$$\rho_o = \frac{1 - \sqrt{k'}}{2(1 + \sqrt{k'})} \quad \rho \approx \rho_o$$

$$0.25 = -20 \log_{10} \left(\frac{1}{\sqrt{1 + \epsilon^2}} \right)$$

$$0.97^2 = \frac{1}{1 + \epsilon^2} \Rightarrow 0.944 + 0.944 \epsilon^2 = 1$$

$$50 = -20 \log_{10} \left(\frac{1}{A} \right)$$

$$\epsilon^2 = 0.059$$

$$k = \frac{\Omega_p}{\Omega_s} = \frac{1}{1.4} = 0.714$$

$$A = 316$$

$$k_1 = \frac{0.242}{\sqrt{316^2 - 1}} = 0.0007658$$

$$N = \frac{2 \log_{10}(4/0.0007658)}{\log_{10}(1/0.0444)} = 5.50$$

$$k' = 0.7$$

$$\rho_o = \frac{1 - \sqrt{k'}}{2(1 + \sqrt{k'})} = 0.0444$$

$$N = 6$$

Ex: butterworth analog HPF

$$\hat{\Omega}_p = 7.5 \text{ k} \quad \hat{\Omega}_s = 1.5 \text{ k} \quad \alpha = 0.25 \quad A = 50 \text{ dB}$$

$$\epsilon^2 = 0.59 \quad A = 316$$

$$\textcircled{1} \quad \Omega_s = \frac{\Omega_p \hat{\Omega}_p}{\hat{\Omega}_s} = \frac{(14)(7.5 \text{ k})}{(1.5 \text{ k})} \Rightarrow \Omega_s = 5 \text{ k}$$

$$\textcircled{2} \quad \frac{1}{k} = \frac{\Omega_s}{\Omega_p} = \frac{5}{1} = 5 \quad k_1 = 0.0007658$$

$$N = \frac{\log_{10}(1/0.0007658)}{\log_{10}(5)} = 4.457 \Rightarrow N = 5$$